Energy Shocks and the Green Adoption Channel in a Small Open Economy

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Abstract

We add an extensive-margin brown/green durable adoption choice to the small open economy HANK model of Auclert et al. (2024). Households choose discretely whether to switch to a green durable that insulates them from fossil energy prices, paying a resource cost. We study brown energy *price* and *supply* shocks under the two fiscal responses studied in the recent literature: a retail price cap (Langot et al., 2026) and a Slutsky-compensating transfer indexed to pre-crisis energy consumption (Bayer et al., 2026). The price cap stabilises output but mechanically eliminates the green adoption margin, while the transfer delivers comparable stabilisation and leaves adoption intact. Shielding consumers from the relative price is therefore not neutral for the energy transition. The transition loss is linear in the cap's intensity under a price shock and convex under a supply shock, and the *sign* of the cap's welfare effect flips with the elasticity of energy supply — reconciling Langot et al. (2026) and Bayer et al. (2026) within a single model. The policy *ordering* is robust to calibration; the *magnitude* of the adoption channel, and even the sign of its output effect under a price shock, depend on the calibrated switching cost and taste scale, which we report and discuss.

1 Introduction

Three recent papers study fiscal responses to the 2022 European energy crisis in HANK frameworks. Auclert et al. (2024) study a small open economy and stress the negative externality when all importers subsidise simultaneously. Bayer et al. (2026) use a two-country HANK with *fixed* euro-area energy supply and find subsidies are beggar-thy-neighbour while Slutsky transfers are not. Langot et al. (2026) evaluate the French *bouclier tarifaire* assuming *perfectly elastic* energy supply and find the price cap performs well. All three treat the household's energy technology as fixed.

We relax exactly that margin. Households can adopt a green durable that insulates them from the fossil price, at a resource cost. The energy shock raises the return to adoption, so the crisis itself can accelerate the transition — unless policy removes the price signal that drives it. Our energy-market closure nests the disagreement between Bayer et al. (2026) and Langot et al. (2026) in a single parameter: with $\varepsilon^E = \infty$ the economy is a price taker (elastic supply, Langot); with ε^E finite the quantity is fixed and the world price clears the market (Bayer). Both closures are reported below.

2 Model

2.1 Environment

The aggregate structure is Auclert et al. (2024): a small open economy with sticky wages, a CES consumption basket nesting energy against a home/foreign bundle, imported energy and foreign goods with sticky retail prices, and a mutual fund holding claims on domestic firms, importers and a share ζ^E of the energy endowment.

Unit of account. The household block is numeraire-generic: it reads a price of the unit of account relative to the CPI, p^{num}, a real return $1 + r^{num} = (1+r)\, p^{num}_{-1}/p^{num}$, and after-tax labour income $atw^{num} = atw/p^{num}$, and returns asset holdings converted by $A^{CPI} = A\, p^{num}$ before meeting CPI-denominated objects in asset market clearing and the current account. We set $p^{num} = p_H$, the domestic good. This makes the resource constraint independent of how the CPI weights brown and green energy, which matters once the two coexist (Section 2.5). Everything contractually written in CPI units — fiscal transfers, the ARS subsistence term, and the switching cost ψ_g — is divided by p^{num} on the way into the budget.

2.2 The durable state

Each household carries a durable state d, the *pair* (holding entering the period, holding chosen last period), needed to charge the switching cost:

$$d \in \{BB,\ BG,\ GB,\ GG\},$$

where GB (green now, brown before) is the state that pays ψ_g. Brown is absorbing; green breaks down to brown at rate δ_g. The state BG carries zero mass by construction, so the effective state is three-point. Stages, in forward order, are `[dep, prod, durables, consav]`: breakdown, idiosyncratic productivity, the discrete adoption choice (a logit with taste scale σ_ε), and the EGM consumption–savings step. The adoption stage uses the discrete-continuous structure of Iskhakov et al. (2017).

2.3 Energy prices

We distinguish three prices, all relative to the CPI:

p^E wholesale/retail market price of brown energy, before any subsidy	
$P^E_B = (1-\tau^E)\, p^E + \tau^E \bar{p}^E$	price paid by a *brown* household
$P^E_G = \varrho\, \bar{p}^E\, \left(P^E_B/\bar{P}^E_B\right)^{\rho_g}$	price paid by a *green* adopter

with ϱ the steady-state green/brown ratio and ρ_g the pass-through of the fossil shock to the green price.

2.4 How the durable enters the budget

Households of different durable types face different energy prices, hence different *consumption bundle prices*. We give each type its own exact CES cost-of-living index, reusing ARS's own energy-nest parameters (α_E, η_E):

$$p^E_d = P^E_B + \mathbb{I}[d \text{ green}]\,(P^E_G - P^E_B), \qquad p_d = \Big[1 + \alpha_E\big((p^E_d)^{1-\eta_E} - (P^E_B)^{1-\eta_E}\big)\Big]^{\frac{1}{1-\eta_E}},$$

and the budget constraint is $(p_d/p^{num})\, c + a' = \mathrm{coh}(d)$. The second expression is exact given the CES identity enforced by the equilibrium block, so brown households have $p_d \equiv 1$ at every date. With type-specific prices, aggregate energy demand is $\sum_d c_E(d)$, not the CES demand evaluated at the aggregate index — the two differ by Jensen's inequality (0.5% here) — so aggregate c_H, c_F, c_E are built from the household block's own heterogeneous-agent outputs.

2.5 What the CPI measures

With two energy prices in the economy, the energy leg of the CPI could be anchored on the brown price alone or on the share-weighted index $[(1-D^G)(P^E_B)^{1-\eta_E} + D^G(P^E_G)^{1-\eta_E}]$. We anchor on the brown price: it is what statistical agencies do (the energy basket is not reweighted quarterly for technology adoption), it preserves $p_d \equiv 1$ for brown households exactly, and feeding D^G back into the CPI would close a DAG cycle requiring the CPI to be promoted to a model unknown. Because the unit of account is the domestic good, this choice does not touch the resource constraint; it affects only the measured deflator and, through it, the monetary rule. The gap between the share-weighted and brown-anchored CPI has closed form

$$\phi^{1-\eta_E} = 1 + \alpha_E\, D^G \Big[(P^E_G)^{1-\eta_E} - (P^E_B)^{1-\eta_E} \Big],$$

computed ex post outside the model. On the price shock without policy it reaches 0.14 annualised percentage points, about 1.5% of the amplitude of measured CPI inflation (0.23 pp, 2.3%, on the supply shock; Figure 1).

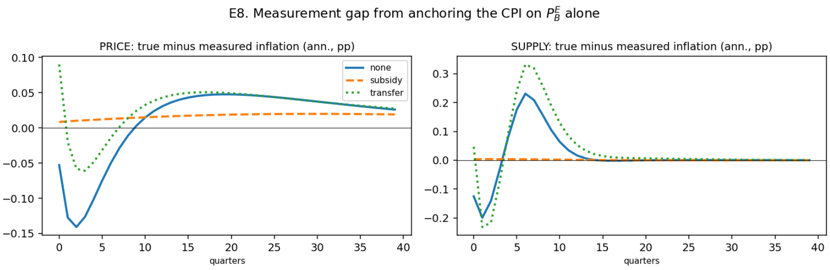

Figure 1: Measurement gap from anchoring the CPI on P^E_B alone: true minus measured annualised inflation, computed ex post.

2.6 Balance of payments

Two durable-margin terms enter the current account. The **switching cost** is booked as an import, $\text{imports}_{dur} = \psi_g D^{sw}$: booking it as domestic demand instead fails here, since with $\mu_{ss} > 1$ and capitalised profits, extra demand for the home good creates a marginal dividend owned by nobody. The **energy saving** of adopters, $(P^E_B - P^E_G)\, C^G_E$, is booked as a real import saving: this asserts that cheaper energy for adopters is a genuine resource saving for the country, not a domestic transfer, and gives the windfall an external counterpart consistent with asset market clearing.

2.7 Policy instruments

Both instruments are the extreme (full-compensation) versions, as in Bayer et al. (2026).

$$\text{Price cap:} \quad P^E_{B,t} = (1-\tau^E)p^E_t + \tau^E \bar{p}^E, \quad \tau^E = 1 \Rightarrow \text{retail price fixed at pre-crisis level} \tag{1}$$

$$\text{Slutsky transfer:} \quad tr_{it} = \iota\,(p^E_t - \bar{p}^E)\,\bar{c}_{E,i}, \qquad \iota = 1 \Rightarrow \text{full compensation on } \textit{pre-crisis} \text{ use} \tag{2}$$

Both are debt financed and repaid through the distortionary labour tax. The transfer base $\bar{c}_{E,i}$ is each household's steady-state energy demand. The cap operates directly on P^E_B, exactly the price entering the brown–green gap, and therefore compresses the adoption incentive; the transfer is lump-sum with respect to current choices and leaves the relative price intact.

3 Calibration

Parameter	Value	Source / target
α_E energy expenditure share	0.04	Auclert et al. (2024)
η_E energy substitution	0.10	Auclert et al. (2024)
r, μ_{ss}	0.01, 1.03	Auclert et al. (2024)
β heterogeneity (3 types)	spread 0.06	MPC ≈ 0.30
δ_g green breakdown rate	0.05	assumption (5 yr life)
$\varrho = P_G^E/P_B^E$ steady-state gap	0.80	assumption
ψ_g switching cost	0.538	targets $D_{ss}^G = 0.05$
σ_ε taste scale	0.05	calibrated jointly with ψ_g
ρ_g pass-through to green price	0	upper bound on the channel

Table 1: Calibration. The steady state is identical across all experiments. ψ_g and σ_ε are calibrated jointly to the level target $D_{ss}^G = 0.05$; σ_ε is not separately pinned down by an adoption-elasticity moment (Section 6).

4 Experiments and results

The *price* shock is a 100 log-point rise in the world energy price with a 16-quarter half-life, as in Auclert et al. (2024). The *supply* shock is a 5% cut in available energy for 6 quarters, after which the world price clears the market endogenously. Impulse responses are reported as 100× the level deviation from steady state: a percent deviation for variables normalised to one (y, C, n, w, prices), percentage points for population shares (D^G, D^{sw}), and a share of the energy steady state (≈ 0.040) for energy quantities.

Shock	Policy	$y(0)$%	$\sum y$	$\pi(0)$ ann.	peak D^G pp	fiscal cost	adoption $\to \sum y$
price	none	−1.12	−30.1	9.14	**9.18**	0.00	−1.60
price	cap	−0.00	−20.9	−1.53	**0.01**	53.82	+0.36
price	transfer	+1.20	+12.7	11.08	**9.32**	53.39	−1.74
supply	none	−0.71	−8.16	10.06	**2.33**	0.00	+0.38
supply	cap	−0.74	−46.9	−0.96	**0.01**	26.09	+0.18
supply	transfer	+1.33	+2.72	14.46	**3.39**	26.56	+2.48

Table 2: Cumulative sums over 24 quarters. The last column is the contribution of the adoption margin: the difference between the model with the margin open and the same model with it shut.

4.1 The crisis moves the transition, in a shock-dependent direction

With no policy, the green share rises by up to 9.2 percentage points under the price shock (from 5.0% to 14.2% of households) and 2.3 pp under the supply shock (Figure 2). The output effect of the adoption margin, however, is not one-signed. Under the **supply** shock it is expansionary (+0.38 points of cumulative output with no policy, +2.48 under the transfer): adopters escape the energy bill and their real income falls less. Under the **price** shock it is contractionary (−1.60 points with no policy, −1.74 under the transfer): the switching cost is booked as an import in exactly the periods household income is falling hardest (Section 2.6), and at the calibrated ψ_g this resource cost outweighs the energy-bill saving. The sign of the adoption channel's contribution to output is therefore not a general property of the model — it depends on the calibrated switching cost and on which shock is driving adoption.

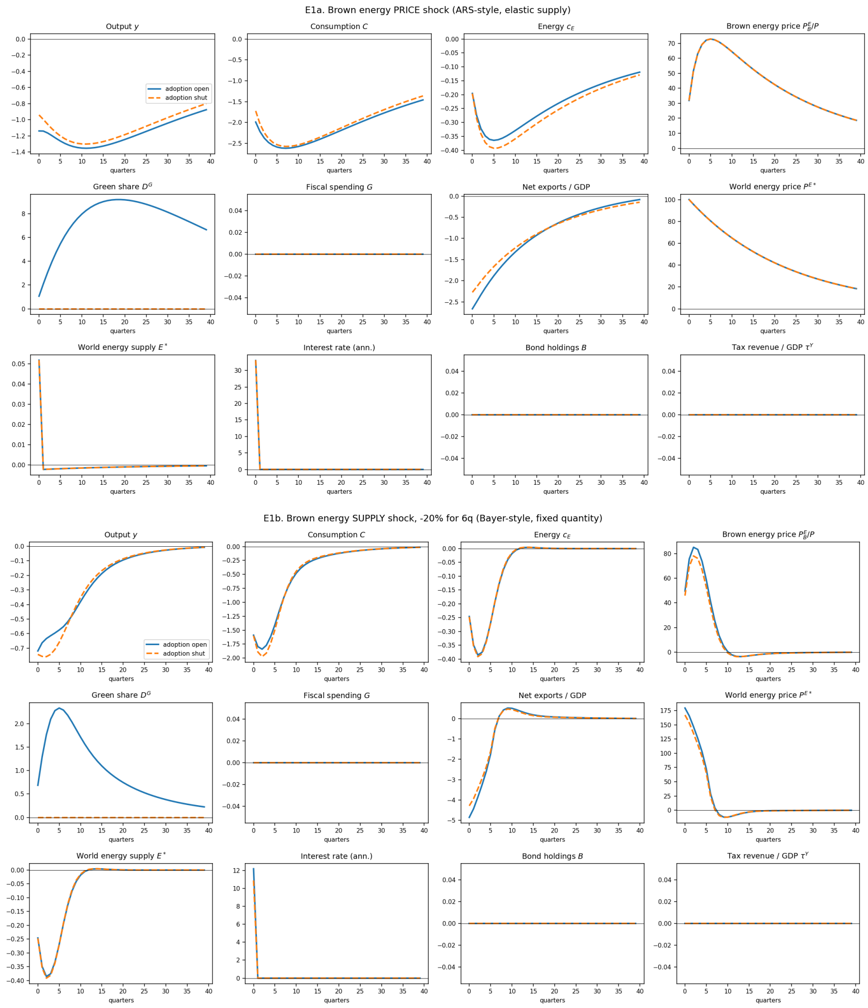

Figure 2: Brown energy shocks, adoption open (solid) vs. shut (dashed), no policy. Top: price shock, elastic energy supply. Bottom: supply shock, -5% for 6 quarters, fixed quantity.

4.2 The price cap eliminates the adoption margin

Under the cap, the peak green share response falls to essentially zero under both shocks (from $+9.18$ to $+0.01$ pp on the price shock, from $+2.33$ to $+0.01$ on the supply shock): the adoption channel is not attenuated, it is switched off. Its contribution to cumulative output collapses to near zero accordingly ($-1.60 \to +0.36$ on the price shock, $+0.38 \to +0.18$ on the supply shock). The mechanism is direct: the cap holds P_B^E at its pre-crisis level, and P_B^E is precisely what determines the return to adoption.

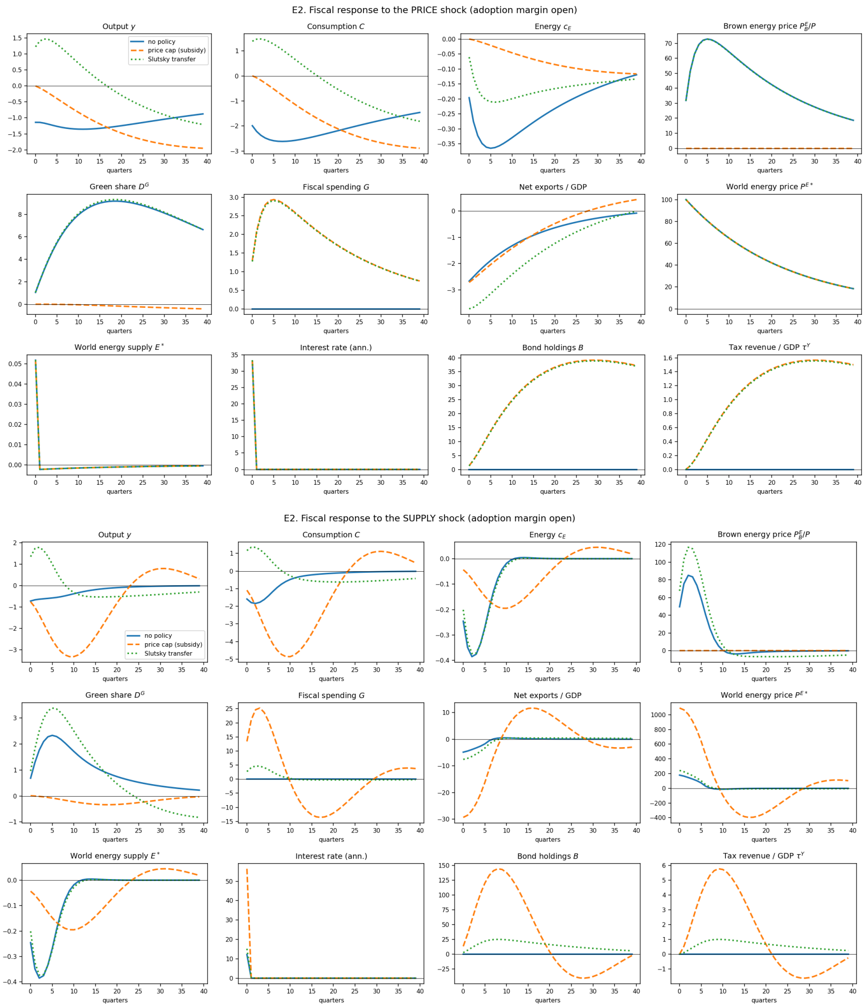

Figure 3: Fiscal response, adoption margin open: no policy (solid), price cap (dashed), Slutsky transfer (dotted). Top: price shock. Bottom: supply shock.

4.3 The transfer preserves adoption, at a cost to output

The Slutsky transfer leaves the green share response essentially unchanged (9.32 vs. 9.18 pp under the price shock, 3.39 vs. 2.33 under the supply shock) while delivering more output stabilisation than the cap, at similar fiscal cost: subsidies destroy the transition margin, transfers do not. Leaving adoption intact is not free for output under the price shock, however: the transfer's cumulative-output gain is *smaller* with adoption open than shut (−1.74 points from the margin, Table 2), for the same resource-cost reason as above.

4.4 Fixed supply makes the cap destructive

Under the supply shock the cap yields cumulative output of −46.9 against −8.2 with no policy: with inelastic supply, capping the price prevents substitution away from energy and pushes the economy towards Leontief. This survives the addition of an adoption margin. It is also precisely the configuration in which Langot et al. (2026), assuming elastic supply, find the cap effective: our elastic-supply column agrees with them (−20.9 vs. −30.1 for cap vs. no policy under the price shock).

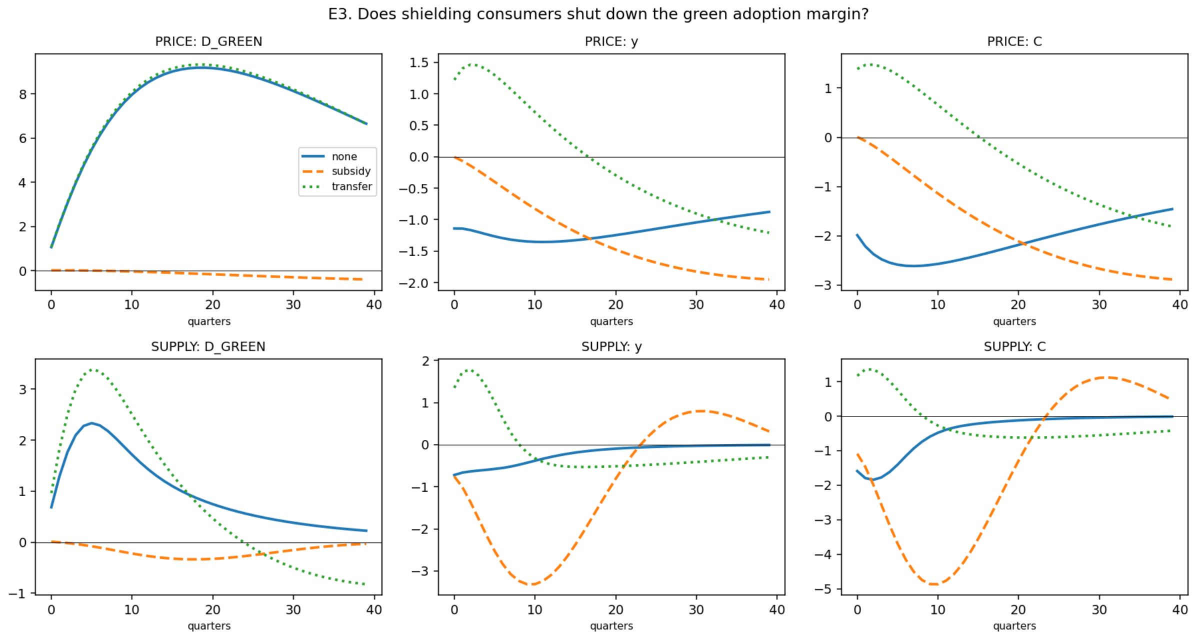

Figure 4: Policy and the adoption margin. Under both shocks the price cap (dashed) flattens the green share response to zero; the transfer (dotted) leaves it intact.

4.5 The cap is a dial, not a switch

Table 2 used $\tau^E = 1$, a complete price freeze. Table 3 sweeps $\tau^E \in [0, 1]$. Under the *price* shock the adoption loss is almost exactly linear in τ^E (half a cap costs a bit under half the transition: 4.53 against 9.18 pp). Under the *supply* shock it is convex: small caps barely touch adoption and the collapse is concentrated near the full freeze. A partial shield is therefore much less damaging to the transition when supply is inelastic — but, as the next result shows, much more damaging to everything else.

Shock	Policy	peak D^G pp	$\sum y$	$\pi(0)$	fiscal	CEV %
price	cap $\tau^E = 0$	9.18	-30.1	9.14	0.00	-1.30
price	cap $\tau^E = 0.25$	6.86	-27.8	6.50	13.39	-1.12
price	cap $\tau^E = 0.50$	4.53	-25.5	3.84	26.83	-0.94
price	cap $\tau^E = 0.75$	2.20	-23.2	1.17	40.30	-0.76
price	cap $\tau^E = 1$	0.01	-20.9	-1.53	53.82	-0.57
price	transfer	9.32	$+12.7$	11.08	53.39	-0.21
supply	cap $\tau^E = 0$	2.33	-8.16	10.06	0.00	-0.59
supply	cap $\tau^E = 0.25$	2.21	-10.60	9.28	5.96	-0.63
supply	cap $\tau^E = 0.50$	1.99	-14.80	8.07	15.34	-0.70
supply	cap $\tau^E = 0.75$	1.50	-23.27	5.89	30.89	-0.84
supply	cap $\tau^E = 1$	0.01	-46.85	-0.96	26.09	-1.28
supply	transfer	3.39	$+2.72$	14.46	26.56	-0.14

Table 3: Dose-response in the cap intensity. CEV is the consumption-equivalent variation relative to the steady state; negative means households would pay that share of permanent consumption to avoid the scenario.

4.6 The welfare effect of the cap flips sign with the supply elasticity

This is the sharpest result in the paper. Under the price shock, tightening the cap *raises* welfare monotonically, from −1.30% to −0.57%. Under the supply shock it *lowers* welfare monotonically, from −0.59% to −1.28%. Langot et al. (2026) assume elastic supply and conclude the French shield worked; Bayer et al. (2026) assume fixed supply and conclude subsidies are destructive. Both are right within their own closure, and our model reproduces each by moving a single parameter: the disagreement in the literature is not about mechanisms, it is an empirical question about the elasticity of energy supply at the relevant horizon.

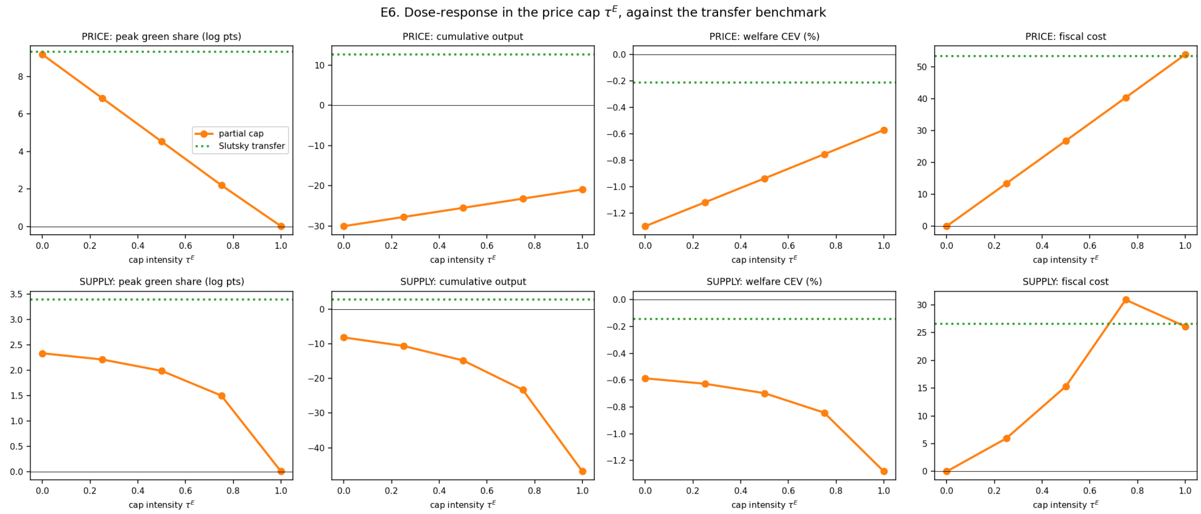

Figure 5: Dose-response in τ^E (orange) against the Slutsky transfer benchmark (dotted). Note the sign reversal of the welfare panel between the two shocks.

4.7 Distributional incidence

Scenario (price shock)	CEV mean	impatient	middle	patient
no policy	−1.30	−2.10	−1.65	−0.14
cap $\tau^E = 0.5$	−0.94	−1.45	−1.25	−0.11
cap $\tau^E = 1$	−0.57	−0.78	−0.85	−0.08
transfer	−0.21	−0.36	−0.45	+0.17

Table 4: CEV by discount-factor type (%). Impatient types hold little wealth and have high MPCs.

Without policy the impatient lose far more than the patient (−2.10 vs. −0.14, roughly 15×). The patient remain the least-hurt group throughout the cap sweep. One clear distributional result: under the transfer, the **patient type's CEV turns positive** (+0.17%) while impatient and middle types remain negative. Indexed to pre-crisis energy use, the Slutsky compensation over-compensates low-MPC, low-energy-exposure households relative to their true loss — a symptom of the fiscal block's lack of investment margin to crowd out the transfer (Section 6).

5 Solution accuracy

Solving the model under the CPI numeraire instead of the domestic good gives a bit-identical steady state and impulse responses agreeing to 2×10^{-5} on y and 3×10^{-5} on C (about 0.15%

of the respective amplitudes) — a relabeling of the same economy, not two different models.

The first-order solution understates the adoption margin at the shock sizes used in this literature. Table 5 compares the fully nonlinear general-equilibrium transition against its linearised counterpart at several price-shock sizes.

Non-linear / linear	y	C	c_E	D^G	π ann.
size 0.125	0.96	0.97	0.94	1.16	0.97
size 0.25	0.91	0.93	0.87	1.33	0.95
size 0.50	0.78	0.82	0.73	1.67	0.92

Table 5: Ratios at the date of the linear peak. Output and inflation are damped relative to the linear prediction while D^G is amplified: a larger share of households escapes the energy bill than the linear solution predicts. The nonlinear solver does not converge at size 1.00, the size used in Table 2 for the price shock.

The 5% supply cut used throughout this paper (rather than the 20% cut used in some of the literature) is motivated by the same non-convergence concern: the first-order solution is not reliable at larger shock sizes, and at some sizes there is no nonlinear benchmark to compare it against at all.

6 Limitations

1. **The taste scale σ_ε is calibrated, not estimated.** ψ_g and σ_ε jointly hit $D^G_{ss} = 0.05$, but one level moment cannot separately identify both a switching cost and a taste scale: a second moment (e.g. the semi-elasticity of adoption to a purchase subsidy) is needed and would discipline the sign found in Section 4.1 along with the magnitudes throughout. The *ordering* of the cap-vs-transfer results does not depend on it, since it follows from the relative price alone.

2. **The first-order solution understates the adoption margin** at the shock sizes used in this literature (Section 5), which in turn overstates the recession it causes.

3. **Zero pass-through to the green price** ($\rho_g = 0$) means adopters are completely insulated from the fossil shock: an upper bound on the adoption channel. A partial pass-through calibrated to observed electricity–gas co-movement would be more defensible.

4. **No within-country energy-share heterogeneity.** Bayer et al. (2026)'s central distributional mechanism is that poor households have higher energy shares; we have homothetic energy demand conditional on the durable type, so we cannot speak to their distributional results.

5. **No investment margin.** Both Bayer et al. (2026) and Auclert et al. (2024) have richer supply sides. Its absence here likely makes the transfer look more expansionary, and the supply shock more inflationary, than it would be with capital to smooth the adjustment.

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